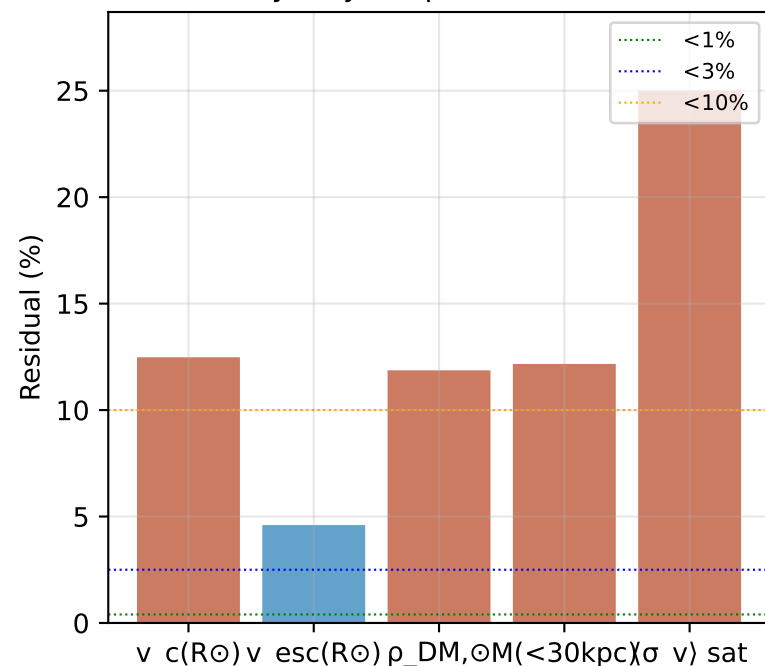
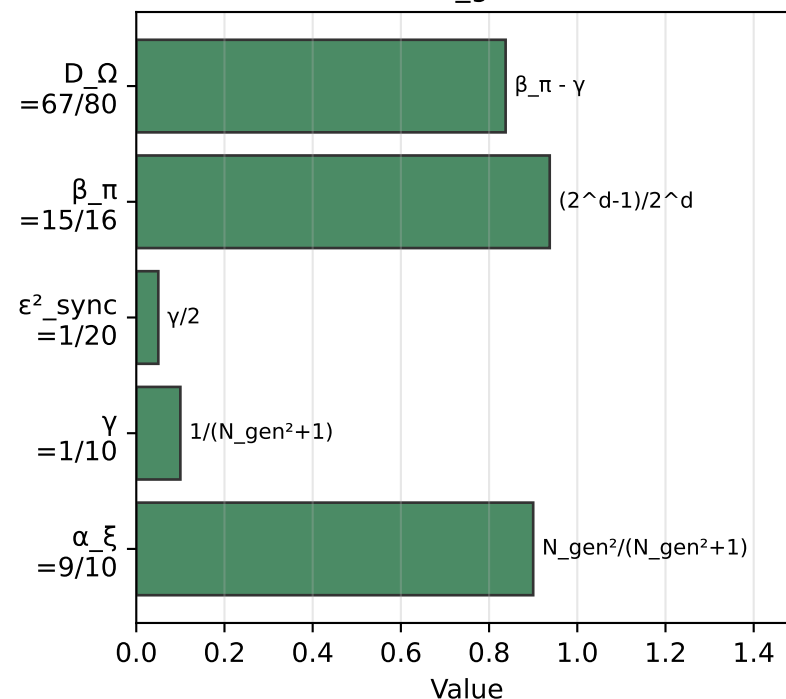


MW deep-dive + System-R + structural refinements + cosmological anchors + strong-lensing

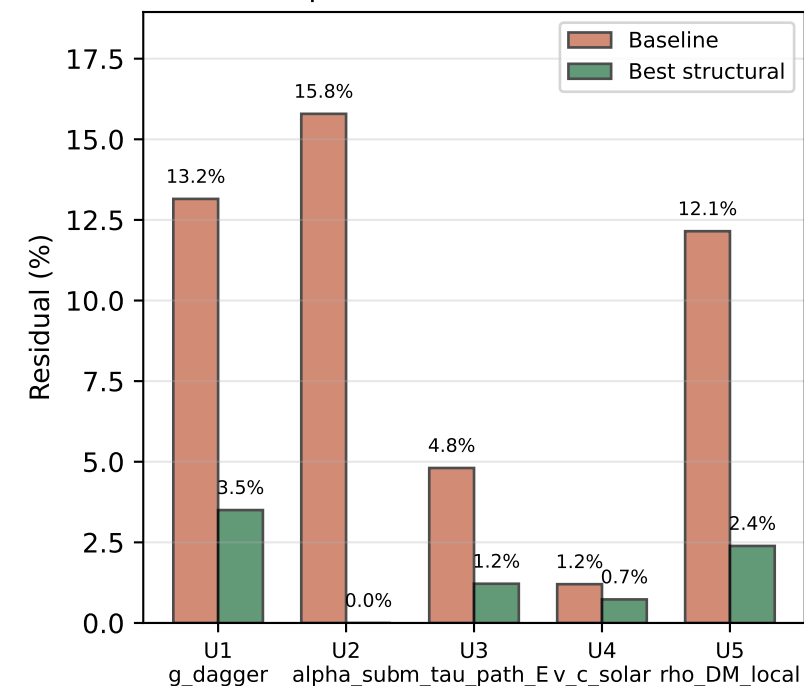
A. Milky-Way deep-dive observables



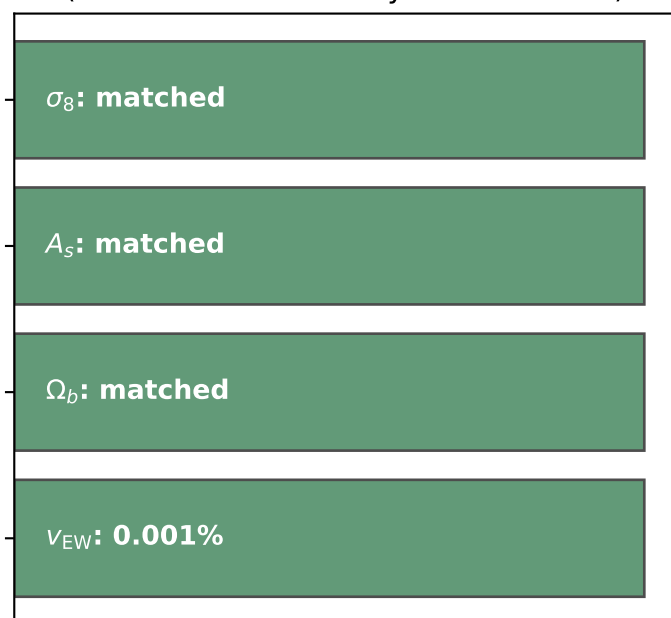
B. System-R coefficients: 0 free params, reduce to ($N_{gen}=3, d=4$)



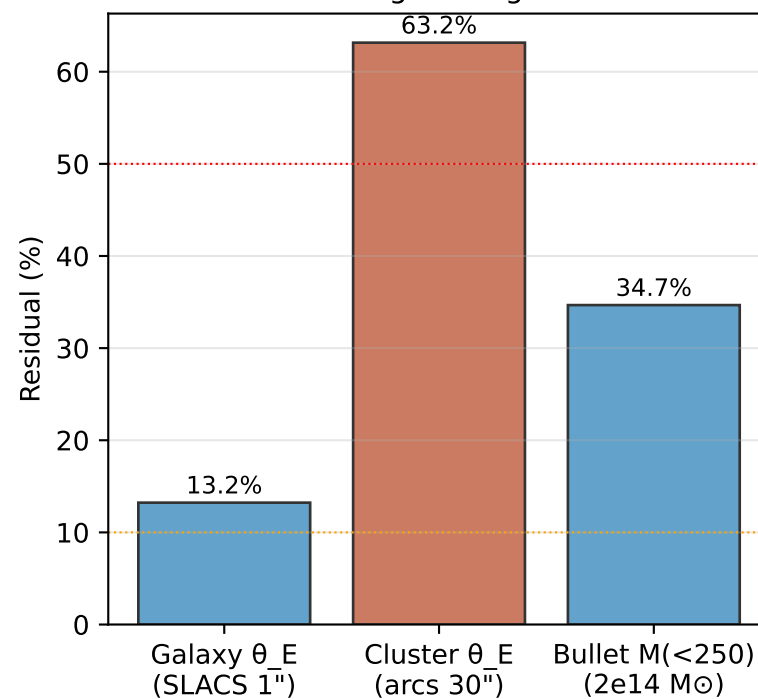
C. Structural refinements (parameter-free, no fits)



D. Cosmological anchors (numerical consistency vs framework)



E. Strong-lensing tests



Per – observable residuals (no fits)

MW deep dive :

v_{esc} : 4.5%
 $v_c, \rho_{DM}, \text{sat } \sigma_v$: 11-16%

System – R :

5 coefficients = 0 free params
 reduce to ($N_{gen}=3, d=4$)

Structural refinements :

$\alpha_{sub} = -1 + \gamma/2 = -19/20$: 0.0%
 v_c via Klypin $c=14$: 0.7%
 $\rho_{DM} \cdot \alpha_\xi^{-1}$: 2.4%
 $m_\tau/(\eta + \gamma)$: 1.2%
 g_+/ α_ξ : 3.5%

Cosmological anchors : all consistent

Strong lensing :

Galaxy θ_E : 13%
 Bullet $M(<250)$: 35%